

SPL1 Project Proposal Form, 2026
Institute of Information Technology (IIT)
University of Dhaka

Student's Name:	G. M. Seyam Chowdhury		
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<u>Project Title: Native Optical Mark Recognition</u>			
<u>Problem Statement and Motivation:</u> Manual checking of OMR answer sheets is time-consuming. The project aims to automatically identify selected answers from OMR sheets while avoiding dependence on third-party image-processing libraries such as OpenCV or Tesseract.			
<u>Proposed Solution and Key Features:</u> A lightweight native OMR scanner will read the image pixel data directly, detect the sheet orientation, correct rotation (when necessary), convert the image to grayscale, apply thresholding and noise reduction, locate answer bubbles, and classify selected answers by comparing filled/dark pixel regions.			
<u>Target Users:</u> Users who need to process and evaluate OMR answer sheets automatically, such as educational or examination-related users.			
<u>Expected Outcome:</u> A working lightweight application that accepts an OMR image, handles incorrect alignment or orientation, detects the answer bubbles, determines the selected option for each question, and produces the corresponding final answer list.			
Languages or Tools to be used: Custom image-processing routines, direct pixel-data processing, grayscale conversion, thresholding, noise reduction, structural/orientation analysis, and bubble detection. The system will not rely on OpenCV or Tesseract.			
Supervisor's Name: Mohd. Zulfiquar Hafiz Jewel			
Signature of the supervisor: _____			
Date: 22-09-2026			

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback:**Before the Final Presentation:**

I confirm that the project is complete and I am forwarding it for final presentation.

Signature of the supervisor: _____

Date: _____